Project: Ubuzima-Link

BSc. in Software Engineering

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CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Healthcare referral systems play a critical role in ensuring that patients receive appropriate medical care at the correct level of the healthcare system. A referral occurs when a healthcare provider transfers a patient to another facility that has the resources, equipment, or specialists required for further diagnosis or treatment. Effective referral systems contribute to better patient outcomes, improved continuity of care, and more efficient use of healthcare resources.

Globally, healthcare systems have increasingly adopted digital technologies to improve patient record management and communication between healthcare facilities. Countries such as Australia, Canada, India, and the United Kingdom have implemented electronic health record systems that support information sharing and care coordination across multiple healthcare providers. These systems help reduce paperwork, improve access to patient information, and support timely clinical decision-making.

Despite these advances, many healthcare facilities in low and middle income countries continue to face challenges related to internet connectivity, limited infrastructure, and fragmented patient information systems. According to the World Health Organization (WHO, 2021), strengthening digital health systems is essential for achieving Universal Health Coverage and Sustainable Development Goal 3 (Good Health and Wellbeing). However, healthcare facilities in resource constrained environments often struggle to benefit fully from digital health technologies because many systems depend on stable internet connectivity.

In Rwanda, the government has invested significantly in digital health through initiatives such as electronic medical records and national health information systems. These efforts have improved healthcare service delivery and health data management. Nevertheless, referral processes between health posts, health centres, and district hospitals still face operational challenges, particularly in facilities where internet access is unreliable or unavailable.

Many referrals continue to rely on paper based documentation, which can be lost, delayed, or damaged during patient transfers. In addition, communication between referring facilities and receiving hospitals is often limited, making it difficult to track referral outcomes and support patient follow up after treatment.

To address these challenges, this project proposes the development of Ubuzima-Link, an offline first referral management system designed to support healthcare facilities operating in low connectivity environments. The system will enable healthcare workers to create, share, and access referral information using local data storage, QR-code based referral sharing, and automatic synchronization when internet connectivity becomes available.

1.2 Problem statement

Healthcare referral systems are essential for ensuring that patients receive timely and appropriate care. However, many healthcare facilities in Rwanda continue to experience challenges in managing referrals efficiently, particularly at the level of health posts and district hospitals.

Although digital health systems have improved healthcare information management in many settings, referral processes in some facilities still depend heavily on paper based records. These records can be misplaced, damaged, or delayed during patient transfers, resulting in incomplete information and disruptions in continuity of care.

Internet connectivity presents an additional challenge. Many digital healthcare systems require continuous access to online services in order to retrieve or update patient records. In facilities where connectivity is unstable, healthcare workers may be unable to access critical patient information when it is needed, forcing them to rely on manual processes.

Furthermore, communication between healthcare facilities is often limited. Referring facilities may not receive timely feedback regarding patient outcomes after treatment, making follow-up care more difficult. As a result, healthcare providers may lack the information needed to monitor patient recovery and coordinate future care.

Existing digital health systems have largely focused on electronic record management and reporting. However, limited attention has been given to referral management solutions that are specifically designed for low connectivity healthcare environments.

Therefore, there is a need for a referral management system that can operate effectively without continuous internet access while supporting secure information sharing and communication between healthcare facilities. This study proposes the development of Ubuzima-Link, an offline first referral management platform that aims to improve referral coordination, information accessibility, and continuity of care between health posts and district hospitals in Rwanda.

1.3 Project's main objective

To design, develop, and evaluate Ubuzima-Link, an offline first referral management system that improves the sharing of patient referral information between health posts and district hospitals in low connectivity environments.

1.3.1 Specific Objectives

1. To identify the challenges affecting referral management between health posts and district hospitals in Rwanda.

2. To design and develop an offline first referral management system using mobile and web technologies.

3. To implement QR-code based referral sharing for secure and efficient transfer of referral information between healthcare facilities.

4. To develop a synchronization mechanism that enables locally stored referral data to be updated when internet connectivity becomes available.

5. To evaluate the usability and effectiveness of the proposed system using prototype testing and feedback from healthcare workers.

1.4 Research Questions

1. What challenges affect referral management between health posts and district hospitals in Rwanda?

2. How can an offline first referral management system improve the sharing of referral information between healthcare facilities?

3. How effective is QR-code based referral sharing in supporting referral processes in low connectivity environments?

4. How usable and acceptable is the proposed system to healthcare workers involved in referral management?

1.5 Project scope

Geographical Scope:
The study will be conducted in Kicukiro District, Kigali, Rwanda. The selected area provides an appropriate environment for evaluating referral management processes between health posts and district level healthcare facilities.

Technical Scope:
The project focuses on the design and development of an offline first referral management system. Key features include:

- Digital referral creation
- QR-code based referral sharing
- Local data storage
- Data synchronization
- Referral status tracking
- Healthcare worker feedback mechanisms

The project will not replace existing national electronic medical record systems but will complement referral management activities between healthcare facilities.

Time Scope:
The project will be carried out over a period of three months, including requirements gathering, system design, development, testing, evaluation, and documentation.

1.6 Significance and Justification

This study addresses a healthcare challenge that affects many low resource settings worldwide, the efficient transfer of patient information between healthcare facilities in environments where internet connectivity is limited.

The proposed system may contribute to improved referral coordination by enabling healthcare workers to access and share referral information more efficiently. By reducing dependence on paper based records, the system has the potential to improve information availability and support continuity of patient care.

For healthcare workers, the system may reduce administrative workload and improve communication between referring and receiving facilities. For patients, improved referral management may contribute to faster access to appropriate healthcare services.

The study also supports Rwanda's ongoing digital transformation efforts in healthcare and aligns with global initiatives aimed at strengthening healthcare systems and improving access to quality healthcare services.

From an academic perspective, the project provides practical experience in software engineering, mobile application development, offline first system design, database synchronization, and healthcare information management.

1.7 Justification of the Study

Although healthcare information systems have improved significantly in recent years, referral management remains a challenge in many low connectivity environments. Existing systems often focus on centralized record management rather than the specific needs of referral communication between healthcare facilities.

The proposed study is justified because it seeks to address an identified gap in healthcare referral management by developing a solution that remains functional even when internet connectivity is unavailable. The study combines software engineering principles with healthcare service needs to create a practical solution that may improve information sharing and continuity of care.

The findings of this study may also contribute to future research on offline-first healthcare applications and referral management systems in developing countries.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Digital health technologies have transformed healthcare delivery by improving the collection, storage, and exchange of patient information. Healthcare organizations around the world increasingly rely on electronic systems to improve communication between facilities, reduce medical errors, and support continuity of care. Among the many functions supported by digital health systems, patient referral management remains one of the most important because it directly affects how patients move between different levels of healthcare services.

The World Health Organization (WHO, 2021) identifies digital health as a key component of strengthening healthcare systems and achieving Universal Health Coverage. However, despite significant progress in healthcare digitization, referral management remains challenging in many low resource settings due to infrastructure limitations, poor connectivity, and fragmented information systems.

This chapter reviews existing literature related to digital health systems, healthcare referral management, offline first technologies, and QR-code applications in healthcare. The chapter also examines international experiences and identifies gaps that justify the development of the proposed Ubuzima-Link system.

2.2 Historical Background of the Research Topic

Digital health refers to the use of information and communication technologies to improve healthcare delivery, patient management, and health system performance (WHO, 2021). Many countries have adopted electronic health records, telemedicine platforms, and mobile health applications to improve healthcare access and efficiency.

In developed countries such as the United Kingdom, Australia, and Canada, electronic health record systems have improved the availability of patient information across healthcare facilities. These systems support coordinated care by allowing healthcare professionals to access patient histories, laboratory results, and treatment records from different locations (Australian Digital Health Agency, 2023).

Similarly, countries such as India have invested heavily in national digital health initiatives. The Ayushman Bharat Digital Mission aims to create a digital infrastructure that supports secure exchange of healthcare information across different providers (National Health Authority, 2023).

Despite these advancements, digital health implementation remains uneven across the world. According to WHO (2021), many healthcare facilities in low and middle income countries continue to face challenges related to infrastructure, funding, workforce capacity, and internet connectivity. These barriers limit the effectiveness of systems that depend entirely on online access.

As a result, researchers increasingly recognize the need for digital health solutions that can operate reliably in environments where connectivity cannot always be guaranteed (Mechael et al., 2022).

2.3 Healthcare Referral Systems

A healthcare referral system is a structured process through which patients are transferred from one healthcare facility to another for specialized diagnosis, treatment, or additional services. Effective referral systems improve healthcare outcomes by ensuring that patients receive appropriate care at the correct level of the healthcare system (WHO, 2021).

Studies conducted in several countries have shown that referral systems often face challenges related to communication, record sharing, and follow up care. In Uganda, Orem et al. (2020) found that delays in referral communication contributed to interruptions in patient treatment and reduced continuity of care. Similar findings have been reported in Tanzania, where incomplete referral documentation affected patient management and follow up services (Kihinga et al., 2021).

In South Africa, electronic referral systems have improved communication between healthcare facilities, but implementation challenges remain due to differences in infrastructure and technology adoption across regions (Leon et al., 2021). These findings suggest that technology alone does not solve referral challenges unless it is adapted to local healthcare environments.

A common challenge identified across many healthcare systems is the lack of effective communication between referring and receiving facilities. Patients are often transferred without complete information, and feedback regarding treatment outcomes may not be communicated back to the referring healthcare provider.

These challenges highlight the need for referral management systems that support efficient information sharing while remaining practical for resource-constrained environments.

2.4 Digital Health and Referral Systems in Africa

- African countries have increasingly adopted digital health technologies to improve healthcare delivery and strengthen health information systems. However, significant disparities remain between urban and rural healthcare facilities.

- Kenya has implemented several digital health initiatives, including electronic medical records and mobile health applications that support healthcare service delivery (Ministry of Health Kenya, 2022). While these systems have improved access to health information, researchers note that internet reliability continues to affect their performance in remote areas.

- In Tanzania, health information systems have improved data collection and reporting processes, but referral coordination remains a challenge in some regions due to infrastructure limitations and inconsistent access to digital technologies (Kihinga et al., 2021).

- Uganda has adopted electronic health information systems to improve patient management and healthcare reporting. Nevertheless, studies indicate that communication between healthcare facilities remains an area requiring further improvement (Orem et al., 2020).

- These experiences demonstrate that although digital health technologies can improve healthcare service delivery, there remains a need for systems specifically designed to function in low connectivity environments.

2.5 Digital Health Systems in Rwanda

Rwanda is recognized as one of Africa's leading countries in digital transformation. The government has invested significantly in information and communication technologies across multiple sectors, including healthcare.

The Ministry of Health has introduced several digital health initiatives aimed at improving patient record management, healthcare reporting, and service delivery (Ministry of Health Rwanda, 2024). Systems such as e-Ubuzima have contributed to improved management of patient information and healthcare data.

Despite these achievements, healthcare facilities continue to experience operational challenges related to referral management. Some facilities still rely on paper based referrals, particularly when internet connectivity is unavailable. This creates difficulties in maintaining complete patient information and tracking referral outcomes.

Furthermore, communication between health posts and district hospitals may be affected by delays in information exchange, making it difficult to support continuity of care after patient transfer.

While Rwanda has made significant progress in digital health implementation, limited research has focused specifically on referral systems that can continue operating during periods of poor connectivity.

2.6 Offline First Systems in Healthcare

Offline first systems are designed to function even when internet connectivity is unavailable. Data is stored locally on the user's device and synchronized with central servers once connectivity is restored.

According to Kleppmann (2017), offline first architectures improve system reliability by reducing dependence on continuous network access. This approach has become increasingly important in healthcare environments where connectivity interruptions are common.

OpenMRS is one example of a health information platform that has been adapted for use in resource constrained settings (OpenMRS, 2024). The system supports healthcare facilities that require flexible approaches to patient data management.

Research conducted in developing countries suggests that offline data collection and synchronization can improve healthcare service continuity and reduce information loss during network outages (Braa et al., 2020). However, many existing systems focus primarily on patient record management rather than referral coordination between healthcare facilities.

This limitation creates an opportunity for systems that combine offline functionality with referral management features.

2.7 QR Codes in Healthcare

Quick Response (QR) codes have become increasingly popular in healthcare because they provide a simple method for storing and retrieving information using mobile devices.

Healthcare organizations use QR codes for patient identification, appointment management, vaccination records, medication tracking, and health certificates (Patel & Singh, 2022). QR codes can improve information accessibility while reducing paperwork and administrative delays.

During the COVID19 pandemic, QR-code based systems were widely used to verify vaccination status and facilitate access to healthcare services in several countries (Aung & Whittaker, 2021).

Although QR-codes provide many benefits, their effectiveness depends on appropriate security measures. Healthcare systems must ensure that patient information remains protected through encryption, authentication, and controlled access mechanisms (HL7 International, 2024).

For referral management, QR codes offer a practical way to transfer referral information between healthcare facilities without requiring constant internet connectivity.

2.8 Review of Existing Systems

Several digital health systems currently support healthcare information management.

1. e-Ubuzima:
e-Ubuzima supports patient record management and healthcare reporting within Rwanda's healthcare system. Its strength lies in national integration and centralized data management. However, some functions may be affected when internet connectivity is unavailable.

2. OpenMRS:
OpenMRS is widely used in developing countries because of its flexibility and open source nature. However, implementation often requires technical expertise and customization, which may limit adoption in smaller healthcare facilities.

3. DHIS2:
DHIS2 is a widely adopted health information management platform that supports data collection, reporting, and monitoring. However, it is not primarily designed for referral management workflows between healthcare facilities.

4. Paper Based Referral Systems:
Paper based systems remain common because they are simple and inexpensive. However, they are vulnerable to information loss, delays, duplication, and difficulties in tracking patient outcomes.

Summary of Existing Systems:
Existing systems provide valuable healthcare information management capabilities. However, most systems either depend heavily on internet connectivity or are not specifically designed to support referral management in low-connectivity environments.

2.9 Research Gap

The reviewed literature demonstrates that digital health systems have significantly improved healthcare information management worldwide. Systems such as e-Ubuzima, OpenMRS, and DHIS2 support patient record management, reporting, and health data exchange.
However, several gaps remain:

First, many existing systems depend on stable internet connectivity, which limits their effectiveness in healthcare facilities operating in low connectivity environments.

Second, much of the available literature focuses on electronic medical records and health information systems rather than referral specific workflows between primary healthcare facilities and hospitals.

Third, limited research has examined the combined use of offline first architecture, local data synchronization, and QR-code based information sharing to support referral management in Rwanda.

As healthcare facilities continue to face challenges related to connectivity, communication, and continuity of care, there is a need for a referral management system that remains functional even when internet access is unavailable.

This study addresses this gap through the development of Ubuzima-Link, an offline first referral management system designed specifically for healthcare facilities operating in low connectivity environments.

2.10 Conclusion

This chapter reviewed literature related to digital health systems, healthcare referral management, offline first technologies, and QR-code applications in healthcare. The review examined international experiences, African healthcare systems, and Rwanda's digital health landscape.

The literature shows that while digital health technologies have improved healthcare service delivery, referral management challenges continue to exist in low connectivity environments. Existing systems provide important healthcare information management functions but do not fully address the specific needs of referral coordination between health posts and district hospitals.

These findings justify the development of Ubuzima-Link as a potential solution for improving referral communication, information accessibility, and continuity of care in Rwanda.

CHAPTER THREE: RESEARCH METHODOLOGY AND SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter describes the methodology used to design, develop, and evaluate the Ubuzima-Link system. It explains the research approach, system development method, data collection techniques, sampling strategy, and ethical considerations. The chapter also presents the system analysis and design, including requirements, architecture, and modelling diagrams.

The purpose of this methodology is to ensure that the system is developed in a structured and systematic way while addressing real challenges faced in healthcare referral management

3.2 Research Design

This study adopts an applied research approach combined with a Design Science Research (DSR) methodology. Applied research is suitable because the project aims to solve a practical problem in healthcare referral management. Design Science Research is appropriate because the study involves building and evaluating a software artifact (Ubuzima-Link).

The research follows an iterative development process, where the system is built in cycles. Each cycle includes design, implementation, testing, and refinement based on feedback.

Component	Description
Research Type	Applied Research
Methodology	Design Science Research
Development Model	Iterative and Incremental Development
Output	Working software system (Ubuzima-Link prototype
Evaluation Method	User feedback and system testing

Table 3.1: Research Design Summary

3.3 Target Population

The target population includes healthcare workers involved in patient referral processes.

Group	Role in Study
Nurses (Health Posts)	Create and send referrals

Clinicians (District Hospitals)	Receive and manage referrals
Health Facility Administrators	Manage facility-level coordination

3.4 Sampling Strategy

The study will use purposive sampling.

Category	Sample Size	Reason
Nurses	5–8	Direct users of referral system
Clinicians	3–5	Handle incoming referrals
Administrators	2-3	Oversee facility operations

Table 3.3: Sampling Plan

3.5 Data Collection Methods

Data will be collected using both primary and secondary sources.

3.5.1 Primary Data

- Interviews with healthcare workers
- Observation of referral processes
- User feedback during prototype testing

3.5.2 Secondary Data

- Academic journals
- WHO reports
- Rwanda Ministry of Health documents
- Existing system documentation (e-Ubuzima, DHIS2, OpenMRS)

3.6 System Development Methodology

The system will be developed using an Iterative and Incremental Development model. Each iteration focuses on improving specific parts of the system.

Phase	Activities
Phase 1	Requirement gathering and analysis
Phase 2	System design (UI, architecture, database)
Phase 3	Core development (offline storage, QR system)
Phase 4	Synchronization and integration
Phase 5	Testing and refinement
Phase 6	Evaluation and documentation

Table 3.4: Development Phases

3.7 System Requirements

3.7.1 Functional Requirements

The system must:

- Allow creation of patient referrals at health posts
- Generate QR codes containing referral identifiers
- Store referral data locally when offline
- Synchronize data when internet becomes available
- Allow hospitals to view and update referral status
- Provide feedback from hospitals to health posts

3.7.2 Non-Functional Requirements:

Category	Requirement
Availability	The system should remain functional during internet outages
Maintainability	The system architecture should support

	future upgrades and modifications
Scalability	The system should support additional healthcare facilities in future deployments
Data Integrity	Referral information should remain accurate during synchronization
Security	Patient data must be encrypted and protected

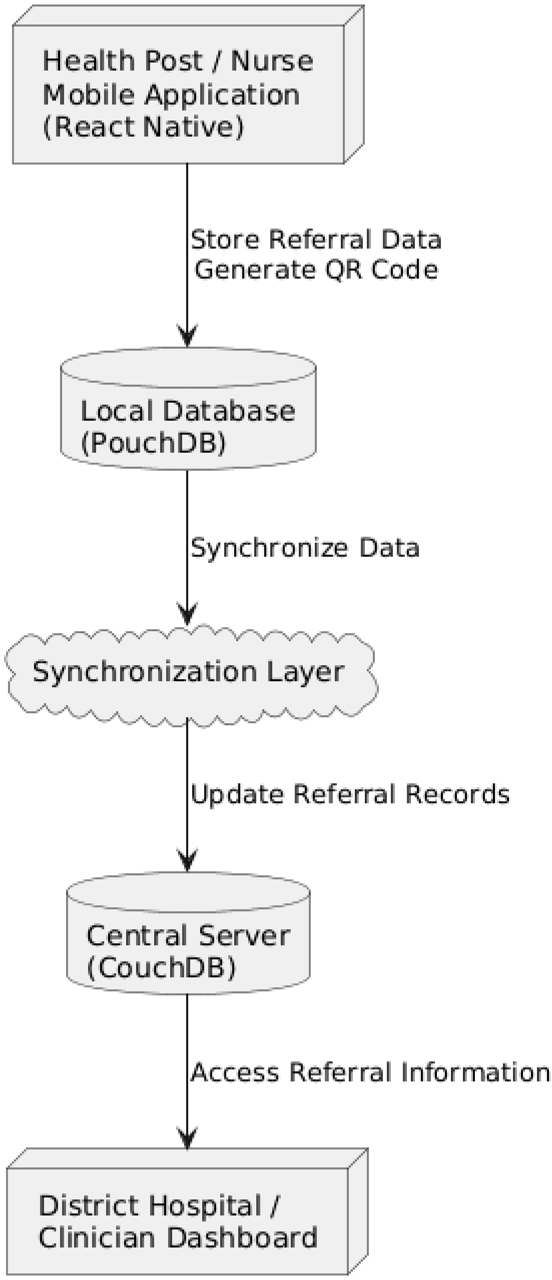
3.8 System Architecture

The system follows an offline first architecture, where data is stored locally and synchronized when connectivity is available.

Description of Components:

- Mobile Application (Health Post): Used to create referrals and generate QR codes
- Local Database: Stores referral data offline using PouchDB
- Synchronization Layer: Handles data sync when internet is available
- Central Server (Hospital System): Stores and manages synchronized referral data

Figure 3.1: System Architecture Diagram



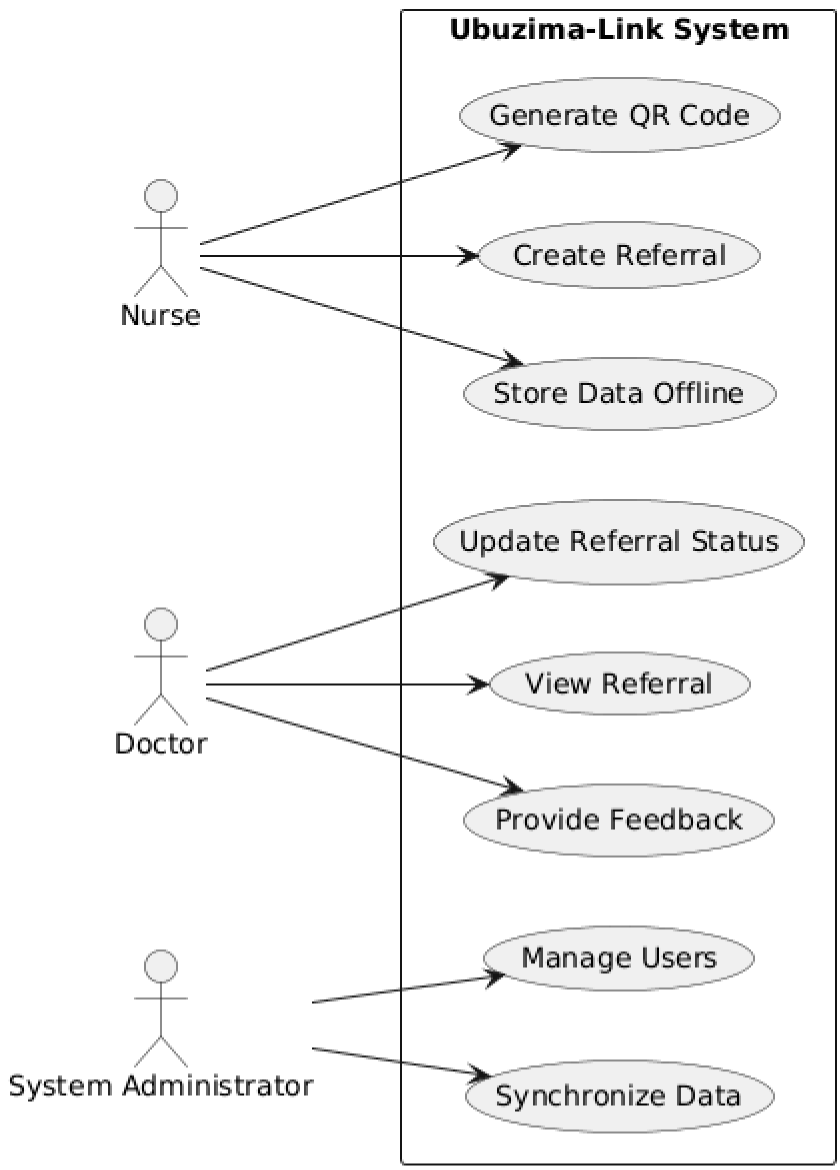
3.9 System Design Models

3.9.1 Use Case Diagram

The system involves three main actors:

- Nurse (creates referrals)
- Doctor (receives and updates referrals)
- System (handles synchronization)

Figure 3.2: Use Case Diagram

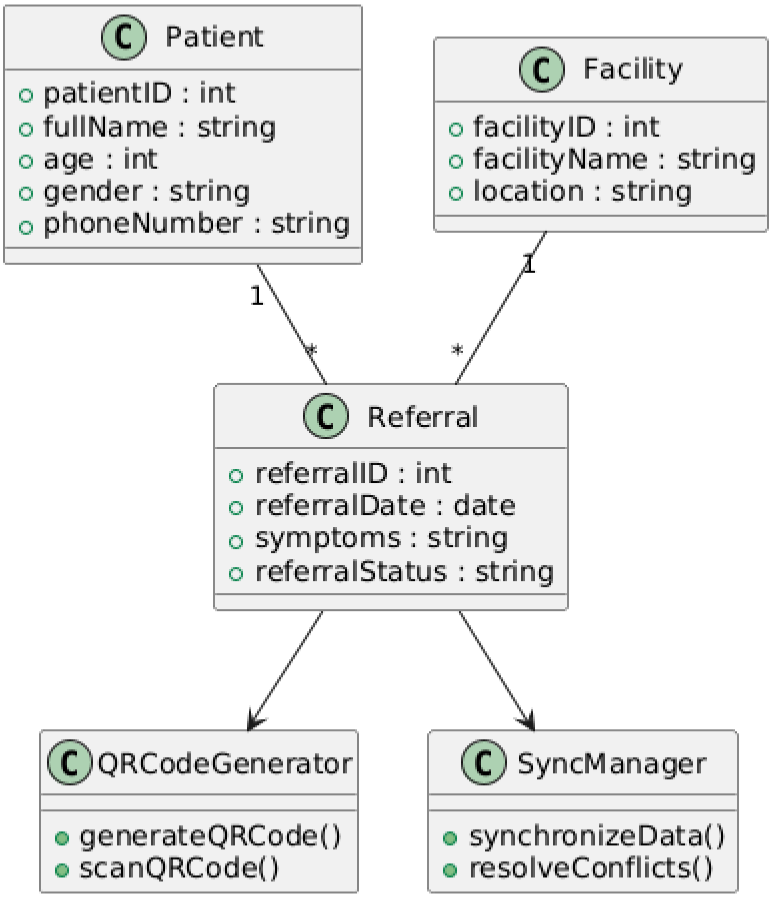


3.9.2 Class Diagram

The system is structured around the following main classes:

- Patient
- Referral
- Facility
- QRCodeGenerator
- SyncManager

Figure 3.3: Class Diagram



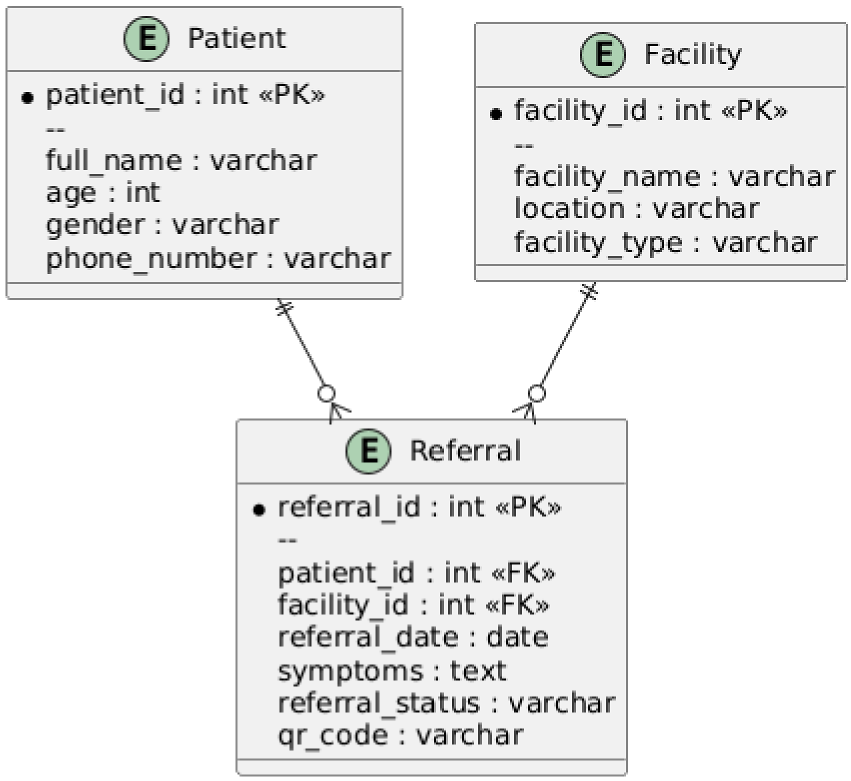
3.9.3 Entity Relationship Diagram (ERD)

The database design ensures proper storage and relationship between entities.

Key relationships:

- One Patient = Many Referrals
- One Facility = Many Patients
- One Referral = One QR Code

Figure 3.4: Entity Relationship Diagram (ERD)



3.10 Tools and Technologies

Category	Tool	Purpose
Mobile Development	React Native	Cross-platform mobile app
Local Database	PouchDB	Offline data storage
Cloud Database	CouchDB	Synchronization and central storage
QR Generation	QR Libraries	Create and scan referral codes
Design Tools	Draw.io / PlantUML	UML and system diagrams

Table 3.5: Development Tools

3.11 Data Analysis Methods

Data collected during evaluation will be analyzed using:

- Descriptive statistics (e.g., frequency of successful referrals)
- Qualitative analysis of user feedback
- Comparison of paper-based vs digital referral time

The evaluation will focus on:
- Referral processing time
- Data completeness
- User satisfaction

3.12 Ethical Considerations:

This study will follow ethical principles in software and healthcare research:

- Participants will provide informed consent before involvement
- No real patient data will be used during testing
- All collected data will be kept confidential
- Participation will be voluntary
- Data will be stored securely with restricted access
- The system will comply with basic healthcare data protection principles

3.13 Summary

This chapter presented the methodology used in developing Ubuzima-Link. It described the research design, sampling strategy, data collection methods, system development approach, and ethical considerations.
The chapter also outlined the system architecture, requirements, and design models that guide the implementation of the system. The next phase of the project focuses on system implementation and testing based on these specifications.

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